Accurate Ensembles, Fragile Narratives: Multi-Scale Stacking and a Fidelity Audit of LLM-Generated Explanations for Credit Risk

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Abstract: Credit scoring increasingly relies on model families whose decision logic cannot be read off their parameters, which places them in tension with supervisory expectations that adverse decisions be explainable. A common proposal is to close that gap with a language model: compute feature attributions, hand them to an LLM, and let it write the rationale. We build such a system end to end and then test whether the second half of the promise holds. The predictive component is a multi-scale stacking ensemble that fuses four differently regularised gradient-boosting learners with a residual network through a small neural meta-learner trained on out-of-fold predictions. On a public 32,581-application credit dataset it reaches a test ROC-AUC of 0.9539 (95% CI $[0.9462, 0.9616]$) and PR-AUC of 0.9137, improving on the strongest single model by $\Delta\mathrm{AUC} = 0.0143$; the improvement remains significant ($p = 0.016$) even under a deliberately conservative independence assumption. The explanation component fuses SHAP and LIME attributions and passes only the top-ranked drivers to a 2.7B-parameter model under greedy decoding. Our central finding is asymmetric. The ranking gains are real but operationally small: at the F_1-optimal threshold the ensemble avoids only six additional missed defaults out of $1{,}422$ relative to a tuned random forest, cutting cost-weighted loss by under 2%. The narrative layer, in contrast, fails in a way that prompt engineering alone does not fix. In an audited case the model named three factors as risk-increasing that the underlying attributions scored as risk-*reducing*. We trace this to properties we measure rather than assume: SHAP and LIME agree on *which* features matter (overlap@10 $= 0.80$) but not on their order ($\tau = 0.43$, $p = 0.18$), and the sign of the attribution for a feature such as applicant income is close to a coin flip across instances (modal-sign share 0.53). We also report calibration (ECS $= 0.117$) and perturbation stability (DPD $= 0.078$) that both fall short of the thresholds our own protocol declares acceptable. We conclude that constrained prompting is a necessary but not sufficient control, and that grounding must be verified after generation rather than assumed from the prompt.

Index Terms: credit risk, stacking ensembles, explainable AI, SHAP, LIME, large language models, explanation fidelity, model calibration

I. Introduction

A credit decision is a decision about a person. When a lender declines an application or prices it punitively, the applicant in most jurisdictions acquires a right to be told why, and the supervisor acquires an interest in whether the stated reason is the operative one [1]. This is what distinguishes credit scoring from most tabular classification problems: the score is not the deliverable. The score plus a defensible account of the score is the deliverable.

The modelling literature has largely optimised the first half. Gradient-boosted trees and their ensembles now dominate credit benchmarks [2], [3], and the margin over interpretable baselines such as regularised logistic regression is large enough that institutions are unwilling to give it up. The standard response is post-hoc attribution: fit whatever model performs best, then explain it with SHAP [4] or LIME [5]. This has become close to a default in credit risk management [6].

Attribution, however, produces a vector of signed real numbers, one per feature, expressed in units of log-odds or probability shift. A loan officer does not read that vector, and neither does an applicant. A recent and fast growing line of work therefore proposes a further translation step: feed the attributions to a large language model and let it write prose [7], [8], [9], [10], [11]. The appeal is obvious. The risk is equally obvious, because language models generate fluent text whether or not it is grounded in what they were given [14], and fluency is precisely what makes an ungrounded credit rationale dangerous: it is more persuasive than the correct one, not less.

Most existing proposals control this risk at the input. They restrict the prompt to a handful of top-ranked drivers, forbid the model from inventing numbers, and decode greedily. We implement exactly this design and then ask the question that the design's advocates usually leave to future work: *does it work?* Not whether the output reads well, since it plainly does, but whether the propositions it asserts match the attributions it was handed.

The answer, on our system, is no. This paper reports that result together with the diagnostic work needed to explain it, and it revises the accompanying predictive claims downward where the evidence requires.

Contributions

1) **A multi-scale stacking ensemble for credit default prediction.** Four gradient-boosting learners deliberately configured at different depth/learning-rate scales, plus a residual network, are fused by a small neural meta-

learner trained on out-of-fold predictions. It attains test ROC-AUC 0.9539 and PR-AUC 0.9137 (Section VII).

2) **Statistical rather than rhetorical model comparison.** We attach confidence intervals to every headline metric and test the ensemble's margin over the best single model under an assumption chosen to disadvantage our own claim. The margin survives ($p = 0.016$).
3) **An operating-point analysis that qualifies the headline.** We show that the ranking gain translates into a cost reduction of under 2% at the deployed threshold across a wide range of misclassification cost ratios (Section VIII). Reporting AUC alone would have overstated the practical benefit.
4) **A measured account of what the attribution layer actually provides.** The ensemble's risk representation differs qualitatively from that of the linear baseline: `loan_grade` absorbs the role that `loan_int_rate` and the bureau default flag play in logistic regression (Section IX). We quantify SHAP–LIME agreement, cross-model agreement, attribution concentration and directional stability instead of asserting them.
5) **A fidelity audit of the narrative layer, with a documented failure.** We define a lightweight sign-and-membership audit, apply it, and report a case in which the generated rationale inverts the direction of three of the four drivers it was given (Section X). We connect this to the attribution instability measured in the previous section rather than treating it as a one-off.
6) **Honest trustworthiness diagnostics.** Calibration, perturbation stability, local explanation consistency and group disparity are reported against the acceptance thresholds our own protocol specifies. Three of the four fail (Section XI). We argue this is more useful to a practitioner than a table of green ticks.

A note on provenance is in order. An earlier version of this work reported a test ROC-AUC of 0.96. Recomputation from the persisted artefacts gives 0.9539. Every figure reported in this paper is recomputed from the serialised experiment outputs, and the reproduction scripts are released with the code.

II. Related Work

A. Machine learning for credit scoring

Benchmarking studies over the past decade converge on a consistent ordering: ensembles of decision trees outperform single learners and outperform neural networks on typical credit-scoring feature sets, with the margin over regularised logistic regression material but not overwhelming [2], [3]. The broader tabular literature reports the same phenomenon and offers explanations grounded in the geometry of tabular decision boundaries: tree ensembles are biased towards axis-aligned, piecewise-constant functions, which suits features such as grade bands and debt-ratio thresholds [19], [20]. Our results reproduce this ordering, with random forest and XGBoost outperforming every neural architecture we test, and our ensemble is designed around it rather than against it, using the neural component as one voice among five rather than as a replacement.

B. Post-hoc attribution and its limits

SHAP [4] grounds attribution in the Shapley value [34], [35], which uniquely satisfies local accuracy, consistency and missingness. LIME [5] fits a sparse linear surrogate in a sampled neighbourhood. Both are model-agnostic and both are widely deployed in finance [6], [36], [37].

Their limitations are also well documented. LIME's reliance on random perturbation makes its explanations unstable across runs [17], and both methods can be deliberately fooled by an adversary who exploits the off-manifold queries they issue [16]. Rudin argues from these failure modes that high-stakes decisions should use models that are interpretable by construction [18]. We do not resolve that debate. We do take from it the methodological point that an attribution vector is evidence about a model, not a transcript of it, and that anything built downstream inherits its noise. Section IX measures that noise on our system; Section X shows it propagating.

C. Language models as explanation interfaces

The proposal to have an LLM verbalise XAI output is recent and active [7], [8]. Kroeger et al. [9] ask whether LLMs can act as post-hoc explainers directly. Factorial studies of narrative quality find that readers prefer LLM-written explanations to raw attribution tables [10], which is the intended effect and also the reason a wrong narrative is worse than a wrong table. Work specific to credit risk asks whether LLMs can serve as governance-grade explainability tools [11] and whether LLM-derived feature rationales align with those of classical models [13]. Two-stage designs that add an explicit verification pass over the generated text [12] are, in our reading, the correct architectural response, and our results are an empirical argument for them.

Faithfulness, meaning whether an explanation reflects the reasoning it purports to describe, has a precise treatment in NLP [15], and hallucination in conditional generation is surveyed in [14]. Our contribution here is narrow and concrete: we apply a minimal faithfulness check to a credit-risk narrative pipeline built the way the literature recommends, and report that it fails.

D. Positioning

Relative to prior work, the novelty of this paper is not the ensemble, and it is not the idea of pairing attribution with an LLM. It is the audit. We are not aware of a credit-risk study that reports both a full attribution-stability profile of its own ensemble and a documented sign-inversion failure of its own narrative layer, and that connects the two.

III. Data and Preprocessing

A. Dataset

We use the public Credit Risk Dataset, comprising $32{,}581$ consumer loan applications described by eleven predictors

TABLE I
PREDICTORS AND TARGET. NAMES ARE GIVEN EXACTLY AS THEY APPEAR IN THE RELEASED DATA AND CODE.

Feature	Description
Numeric predictors	
`person_age`	Applicant age in years
`person_income`	Annual income
`person_emp_length`	Employment tenure in years
`loan_amnt`	Requested principal
`loan_int_rate`	Contractual interest rate
`loan_percent_income`	Principal as a share of income
`cb_person_cred_hist_length`	Length of credit history
Categorical predictors	
`person_home_ownership`	Rent, mortgage, own or other
`loan_intent`	Stated purpose, six levels
`loan_grade`	Internal grade A to G, ordinal, A best
`cb_person_default_on_file`	Prior bureau default, binary
Target	
`loan_status`	1 = default, 0 = repaid

and a binary outcome `loan_status` (1 = default). Seven predictors are numeric and four categorical; Table I lists them. The target is imbalanced: 25,473 non-defaults (78.18%) against 7,108 defaults (21.82%), a ratio of 3.58:1.

Overall completeness is 98.97%. Missingness is confined to two continuous variables, `loan_int_rate` (3,116 records, 9.56%) and `person_emp_length` (895 records, 2.75%), and is scattered rather than structured. Duplicate rows account for 165 records (0.51%).

B. Leakage control

The single most consequential preprocessing decision is ordering. We partition before we learn anything from the data. The design matrix and labels are split 80/20 with `stratify=y`, preserving the 21.82% default rate in both partitions, yielding 26,064 training and 6,517 test records (5,095 non-defaults, 1,422 defaults). Every subsequent transformation (winsorisation caps, imputation, scaling, encoding) estimates its parameters on the training partition only and is then applied unchanged to the test partition. The fitted transformer objects are serialised so that the inference service applies numerically identical transformations.

C. Transformations

Winsorisation. Domain-implausible extremes were capped rather than deleted: `person_age` at 100 years and `person_emp_length` at 50 years, against raw maxima of 144 and 123 respectively. Caps are estimated on training data and reused.

Imputation. The two incomplete numeric columns are filled with a k-nearest-neighbour imputer ($k = 5$). Because the missing variables are correlated with income and principal, local interpolation is preferable to global mean or median substitution.

Scaling. Numeric features are centred on the training median and scaled by the training interquartile range. Income is severely right-skewed (sample skewness ≈ 32, driven by a small number of very high earners), which is exactly the regime in which a robust scaler outperforms standardisation for gradient-based learners.

TABLE II
BASE-LEARNER CONFIGURATIONS. THE FOUR BOOSTING MODELS ARE SPREAD ACROSS THE DEPTH–SHRINKAGE PLANE BY DESIGN RATHER THAN TUNED TO A COMMON OPTIMUM.

Learner	Depth	Trees	η	Subsample	Colsample
XGBoost DEEP	8	500	0.020	0.80	0.80
XGBoost SHALLOW	3	300	0.050	0.90	0.90
LightGBM	6	400	0.030	0.85	0.85
CatBoost	7	450	0.025	0.80	0.80
Residual MLP	n/a	n/a	0.001	n/a	n/a

Encoding. `loan_grade` carries a natural order ($A < \cdots < G$) and is ordinal-encoded so that higher values mean worse credit quality. `person_home_ownership` and `loan_intent` are one-hot encoded with the first level dropped and `handle_unknown='ignore'` so that unseen categories at inference yield a zero vector rather than an exception. `cb_person_default_on_file` is mapped $\{N,Y\} \to \{0, 1\}$.

Concatenation yields a 17-dimensional design matrix used identically by every model reported here.

IV. MULTI-SCALE STACKING ENSEMBLE

A. Design rationale

Stacked generalisation [25] improves on a single learner when the base learners make *different* mistakes. Diversity is therefore the design objective, not accuracy of the individual components. Credit data is mixed-type, moderately imbalanced, and contains both broad monotone trends (higher repayment burden, higher risk) and sharp local interactions (a marginal grade combined with short tenure). We assemble learners whose inductive biases cover that range.

Multi-scale gradient boosting. Rather than four tuned variants of the same configuration, we deliberately spread the learners across the depth–shrinkage plane. A deep XGBoost (depth 8, $\eta = 0.02$, 500 trees) captures high-order interactions. A shallow XGBoost (depth 3, $\eta = 0.05$, 300 trees) acts as a low-variance anchor that represents global monotone structure. LightGBM's leaf-wise growth [22] concentrates capacity on high-gradient regions, improving coverage of sparse segments. CatBoost's ordered boosting [23] reduces target leakage in categorical splits, which matters for the one-hot grade and intent blocks. Hyperparameters are given in Table II.

Residual network. Tree learners approximate smooth functions with staircases. The fifth base learner is a residual MLP over the scaled numeric manifold: dense layers of 512 and 256 units, a residual block expanding back to 512 with an identity skip [26], then 128 and 64 units before a sigmoid output. It contributes a smooth approximator whose errors are, by construction, poorly correlated with the trees'.

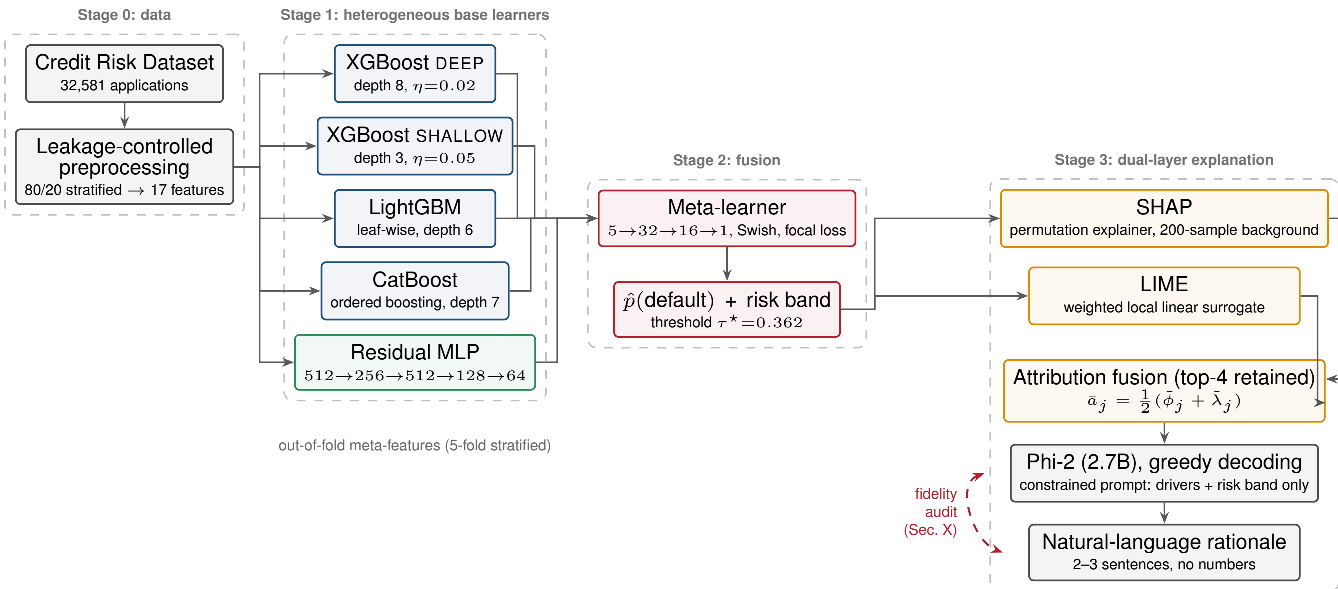

Fig. 1. End-to-end system. Five heterogeneous base learners are fused by a neural meta-learner trained on out-of-fold predictions; the resulting probability drives both the credit decision and the explanation stack. The dashed edge marks the interface audited in Section X, where the attributions handed to the language model are compared against the propositions it asserts.

B. Imbalance handling

Every base learner absorbs the class imbalance before stacking, so that the meta-learner receives comparably scaled signals. The boosting models use

$$w_+ = \frac{1-\pi}{\pi} = \frac{1-0.2182}{0.2182} \approx 3.58, \tag{1}$$

with π the training default rate. The neural components use focal loss [29],

$$\mathcal{L}_{\text{focal}}(p_t) = -\alpha_t (1-p_t)^\gamma \log p_t, \tag{2}$$

with $\alpha = 0.25$ for the positive class, 0.75 otherwise, and $\gamma = 2.0$. Focal loss down-weights the abundant easy negatives and concentrates gradient on the hard positives, which is the behaviour we want when defaults are the minority and the expensive error.

C. Out-of-fold meta-features

Training a meta-learner on in-sample base predictions leaks the target: the base models are overconfident on data they memorised, and the meta-learner learns to trust that overconfidence. We avoid this with the standard out-of-fold construction, illustrated in Fig. 2.

Let $\mathcal{D}_{\text{tr}}$ be partitioned into five stratified folds $\{\mathcal{F}_k\}_{k=1}^5$. For each base learner m and fold k we refit on $\mathcal{D}_{\text{tr}} \setminus \mathcal{F}_k$ and score $\mathcal{F}_k$:

$$Z_{i,m} = \hat{p}_m^{(-k(i))}(\mathbf{x}_i), \qquad i \in \mathcal{F}_{k(i)}, \tag{3}$$

where $k(i)$ is the fold containing sample i. The result is a meta-feature matrix $Z \in \mathbb{R}^{26064\times 5}$ in which every entry is an honest held-out probability. Test meta-features average the five fold-models,

$$Z_{i,m}^{\text{te}} = \frac{1}{5}\sum_{k=1}^{5} \hat{p}_m^{(-k)}(\mathbf{x}_i), \tag{4}$$

which also reduces variance relative to any single refit.

D. Meta-learner

The meta-learner is deliberately small: with only five inputs, capacity is a liability. It maps $5 \to 32 \to 16 \to 1$ with Swish activations [33], L_2 penalty 10^{-4}, batch normalisation [31] after each hidden layer, and dropout 0.2 [32] after the first. Optimisation uses Adam [30] at 10^{-3} with focal loss, plateau-triggered learning-rate halving (factor 0.5, patience 5, floor 10^{-6}) and early stopping on validation ROC-AUC (patience 10, best weights restored). The narrow $32 \to 16$ bottleneck forces compact combination rules; batch normalisation stabilises training on the strongly correlated meta-features.

Monitoring ROC-AUC rather than loss is a deliberate choice under imbalance: validation loss can improve while ranking quality degrades, and ranking is what the downstream threshold consumes.

E. Threshold selection

A 0.5 cut-off is inappropriate at a 21.8% base rate. We select

$$\tau^\star = \arg\max_\tau \frac{2\,\text{Prec}(\tau)\,\text{Rec}(\tau)}{\text{Prec}(\tau)+\text{Rec}(\tau)} \tag{5}$$

on training data, giving $\tau^\star = 0.362$ for the ensemble, and apply it unchanged to the test partition. We stress in Section VIII that F_1 is a convenient but economically arbitrary criterion, and that the choice of τ matters more to realised cost than the choice of model.

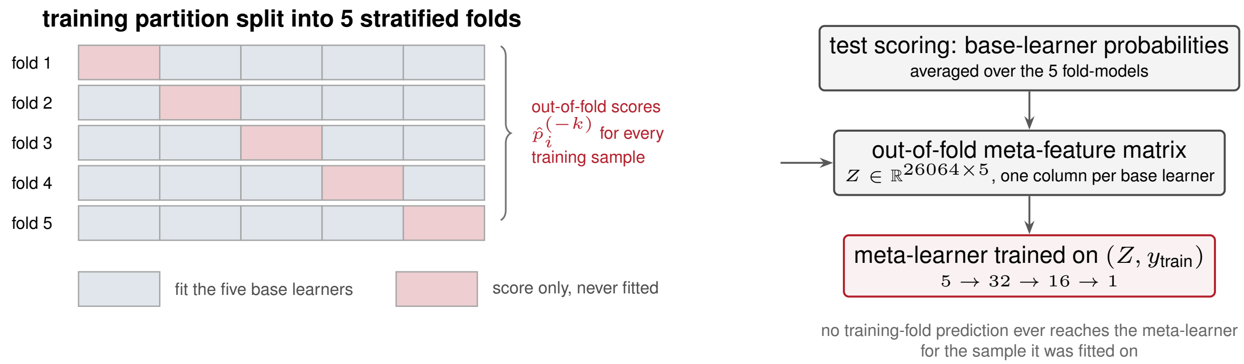

Fig. 2. Out-of-fold construction of the meta-feature matrix. Within each of five stratified folds the base learners are refitted on the complementary four folds and score only the held-out block, so no meta-feature is ever produced by a model that saw the corresponding label. Test-set meta-features average the five fold-models.

V. Dual-Layer Explanation Framework

A. Layer 1: quantitative attribution

SHAP. Because the ensemble is a composition of five heterogeneous learners and a network, no model-specific fast path applies. We wrap the entire pipeline (base scoring, stacking, meta-learner) in a single callable and explain *that*, using a permutation explainer with a 200-sample background drawn from training data. Attributions are computed for 100 test instances, giving $\Phi \in \mathbb{R}^{100\times 17}$. Explaining the composition rather than a component is the methodologically important part: attributions taken from one base learner would describe a model that is not the one making the decision. Global importance is $\bar{\phi}_j = \frac{1}{n}\sum_i |\phi_{ij}|$.

LIME. For each of 50 test instances we fit a proximity-weighted sparse linear surrogate on perturbed neighbours,

$$g^* = \arg\min_{g\in\mathcal{G}} \sum_z \pi_{\mathbf{x}}(z)\big(f(z) - g(z)\big)^2 + \Omega(g),$$
$$\pi_{\mathbf{x}}(z) = \exp\!\big(-\|z - \mathbf{x}\|^2/\sigma^2\big), \tag{6}$$

and take the surrogate coefficients as local importances.

Fusion. SHAP and LIME scores are min–max normalised to a common scale, $\tilde{\phi}_j$ and $\tilde{\lambda}_j$, and averaged, $\bar{a}_j = \frac{1}{2}(\tilde{\phi}_j + \tilde{\lambda}_j)$. The intent is that consensus suppresses method-specific artefacts. Section IX examines whether the two methods agree enough for this averaging to be meaningful.

B. Layer 2: narrative generation

The narrative layer uses `microsoft/phi-2`, a 2.7B-parameter decoder-only transformer [45], loaded in `float32` with left-side padding. Model choice is driven by deployment constraints: narrative generation is an inference-time cost per application, and a 2.7B model runs on the same commodity hardware as the scoring service.

The prompt is deliberately impoverished. It contains the predicted probability, a discretised risk band (HIGH/MODERATE/LOW), the raw applicant profile, the top four fused drivers with signed impacts, the five globally most important features, and the three most sensitive features. It instructs the model to write two to three sentences, to reference at least three named features, to describe how they change risk, and to avoid both jargon and numerals. Decoding is greedy (`do_sample=False`, `repetition_penalty=1.05`). The full template is reproduced in the appendix.

Every control here is an *input-side* control. Nothing in the design inspects the generated text. This is the standard configuration in the literature, and Section X is an argument that it is insufficient.

VI. Experimental Protocol

A. Metrics

Accuracy is excluded. At a 21.8% default rate a constant non-default predictor achieves 78% accuracy while providing no risk discrimination, so accuracy would reward exactly the behaviour we need to detect. We report ROC-AUC for threshold-free ranking quality, PR-AUC for behaviour on the minority class, and F_1, precision, recall and specificity at the selected operating point. Section VIII adds a cost-weighted view.

B. Uncertainty quantification

Point estimates without intervals invite over-reading of small differences. We report a 95% confidence interval for every AUC using the Hanley–McNeil variance approximation [42],

$$\widehat{\mathrm{Var}}(A) = \frac{1}{n_+ n_-}\Big[A(1-A) + (n_+ - 1)(Q_1 - A^2) + (n_- - 1)(Q_2 - A^2)\Big], \tag{7}$$

with $Q_1 = A/(2-A)$ and $Q_2 = 2A^2/(1+A)$, and Wilson score intervals [44] for recall, precision and specificity, which behave correctly near the boundary where the normal approximation does not.

For the ensemble-versus-baseline comparison the correct instrument is a correlated test such as DeLong's [43], which requires the paired score vectors. Those were not retained by the original experiment. We therefore compute the z statistic under the assumption that the two AUCs are *independent*. Since predictions on a shared test set are strongly positively correlated, and $\mathrm{Var}(A_1-A_2) = \sigma_1^2+\sigma_2^2-2\rho\sigma_1\sigma_2$ decreases in ρ, this inflates the standard error and understates significance. Any conclusion that survives it would survive the correct test. We report the sensitivity to ρ explicitly rather than quietly assuming a favourable value.

C. Reproducibility

Experiments ran on Windows 11 with an Intel Core Ultra 9 275HX (24 cores), 32 GB RAM, and GPU acceleration, under Python 3.10.13 with NumPy 1.26, pandas 2.1, SciPy 1.11, scikit-learn 1.4, XGBoost 2.0, LightGBM 4.0, CatBoost 1.2, TensorFlow 2.15, PyTorch 2.0 with pytorch-tabnet 4.0, SHAP 0.45 and LIME 0.2. A fixed seed, 5-fold stratified cross-validation and the 80/20 stratified split are enforced throughout. Training and test tensors, fitted preprocessing objects, trained models and all metric artefacts are serialised; the analyses and figures in this paper are regenerated from those artefacts by released scripts.

VII. Predictive Performance

Table III and Fig. 3 report the full comparison.

The ordering reproduces the tabular literature. Random forest (0.9396) and XGBoost (0.9358) beat every neural architecture we trained. The best network, the residual MLP, reaches 0.9306; TabNet's learnable tokenizer variant is the weakest of the deep models at 0.9197, so the additional feature-tokenisation machinery cost more than it returned on a 17-dimensional design matrix. This is consistent with the finding that tree ensembles retain an advantage on typical tabular data [19], [20]. It is worth stating plainly because it undercuts a common framing: the deep architectures here are not where the performance comes from.

The ensemble improves on the best single model, and the improvement is statistically supported. Against the random forest, $\Delta\mathrm{AUC} = 0.0143$. Under the conservative independence assumption of Section VI, $z = 2.40$ and $p = 0.016$. Under more realistic correlation the margin is far more decisive: at $\rho = 0.5$, $p = 7.3 \times 10^{-4}$; at $\rho = 0.8$, $p = 1.3 \times 10^{-7}$. Against the best neural architecture, $\Delta\mathrm{AUC} = 0.0233$ with $p = 1.6 \times 10^{-4}$ even under the conservative assumption. The stacking gain is therefore real rather than noise, which is the claim H1 makes and the evidence supports.

Overfitting is controlled. The train–test ROC-AUC gap does not exceed 0.0082 in absolute value for any model, and is negative for the ensemble (-0.0056, i.e., test exceeds train). For a system with five base learners and a meta-network this is the reassuring outcome, and it is attributable to the out-of-fold protocol and early stopping rather than to luck.

PR-AUC separates the models more sharply than ROC-AUC. The ensemble's 0.9137 against the random forest's 0.8948 is a 0.0189 gap, larger than the corresponding ROC gap. Under imbalance this is the more relevant comparison, since PR-AUC is sensitive to performance on the minority class that motivates the exercise.

VIII. Operating Point and Cost

Table III reports ranking quality. Deployment consumes decisions, and the translation is not automatic.

Consider the confusion matrices in Table IV. At its selected threshold the ensemble produces 364 false negatives and 40 false positives. The random forest produces 370 and 44. The ensemble's entire operational advantage over the strongest baseline is *six* additional defaults caught and *four* fewer good applicants wrongly declined, out of 6,517 decisions. The Wilson intervals for recall overlap almost completely ($[0.721, 0.766]$ versus $[0.716, 0.762]$).

To make this precise, let c_{FN} and c_{FP} be the costs of a missed default and a wrongly declined applicant, and define expected cost per thousand applications

$$C(\kappa) \;=\; \frac{1000}{n}\bigl(\mathrm{FP} + \kappa\,\mathrm{FN}\bigr), \qquad \kappa = c_{\mathrm{FN}}/c_{\mathrm{FP}}. \tag{8}$$

Fig. 4 plots the excess of each model over the ensemble. At $\kappa = 5$ the ensemble costs 285.4 against the random forest's 290.6, a 1.80% reduction; at $\kappa = 10$, 564.7 against 574.5 (1.71%); at $\kappa = 20$, 1123.2 against 1142.2 (1.67%). The reduction is remarkably stable in κ and remarkably small.

Two conclusions follow, and they pull in opposite directions.

First, the honest one: *a* 0.0143 *AUC gain does not buy* 0.0143 *worth of anything at a fixed threshold.* ROC-AUC integrates over all thresholds; a deployed system occupies one. Papers that report only AUC improvements, including this one as originally drafted, invite the reader to infer an operational benefit that the confusion matrix does not support. We would rather state the limitation than let the abstract carry an implication we cannot defend.

Second, the qualification: all models here are evaluated at *their own* F_1-optimal thresholds, and F_1 weights precision and recall equally, which corresponds to $\kappa \approx 1$. No lender believes that. At $\kappa = 10$, every model in Table IV is operating well away from its cost-optimal point, and the ensemble's superior ranking gives it more room to move: a better-ordered score list yields a better precision–recall frontier to select from. Threshold selection under an explicit cost model is, on this evidence, a higher-leverage intervention than further architecture search, a point we return to in Section XII.

IX. What the Attribution Layer Actually Shows

A. The ensemble's risk representation is not the linear model's

Fig. 5 gives the global profile. `loan_percent_income` leads at $\bar{\phi} = 0.0827$, followed by `loan_grade` (0.0768) and `person_income` (0.0464).

The interesting result is what is *absent*. In the L1-regularised logistic baseline, the largest positive coefficients belong to `loan_percent_income`, `loan_int_rate` and the bureau default flag. In the ensemble, `loan_int_rate` ranks twelfth (0.0131) and `cb_person_default_on_file` sixteenth of seventeen (0.0020).

This is not a contradiction, and it is not a defect. `loan_grade` is an internal rating that is itself a function of interest rate and bureau history: grade determines price.

TABLE III
TEST-SET PERFORMANCE. ALL VALUES RECOMPUTED FROM THE SERIALISED EXPERIMENT ARTEFACTS. AUC INTERVALS ARE HANLEY–MCNEIL; $\tau^\star$ IS THE F_1-OPTIMAL THRESHOLD SELECTED ON TRAINING DATA. THE GENERALISATION GAP COLUMN IS TRAIN MINUS TEST ROC-AUC.

Model	ROC-AUC	95% CI	PR-AUC	F_1	Precision	Recall	Specificity	Gap
Multi-scale ensemble	**0.9539**	[0.9462, 0.9616]	**0.9137**	**0.8397**	0.9636	**0.7440**	0.9921	−0.0056
Random forest	0.9396	[0.9309, 0.9484]	0.8948	0.8356	0.9599	0.7398	0.9914	−0.0082
XGBoost	0.9358	[0.9268, 0.9448]	0.8867	0.8241	**0.9787**	0.7117	**0.9957**	−0.0079
Residual network	0.9306	[0.9213, 0.9399]	0.8764	0.8094	0.9273	0.7180	0.9843	+0.0063
Deep & Cross network	0.9251	[0.9155, 0.9348]	0.8653	0.7933	0.9076	0.7046	0.9800	−0.0027
TabNet (pure)	0.9218	[0.9120, 0.9317]	0.8608	0.7946	0.9180	0.7004	0.9825	+0.0051
TabNet + tokenizer	0.9197	[0.9098, 0.9297]	0.8462	0.7747	0.8787	0.6927	0.9733	+0.0012
Logistic regression	0.8663	[0.8538, 0.8787]	0.6956	0.6454	0.6248	0.6674	0.8881	−0.0027

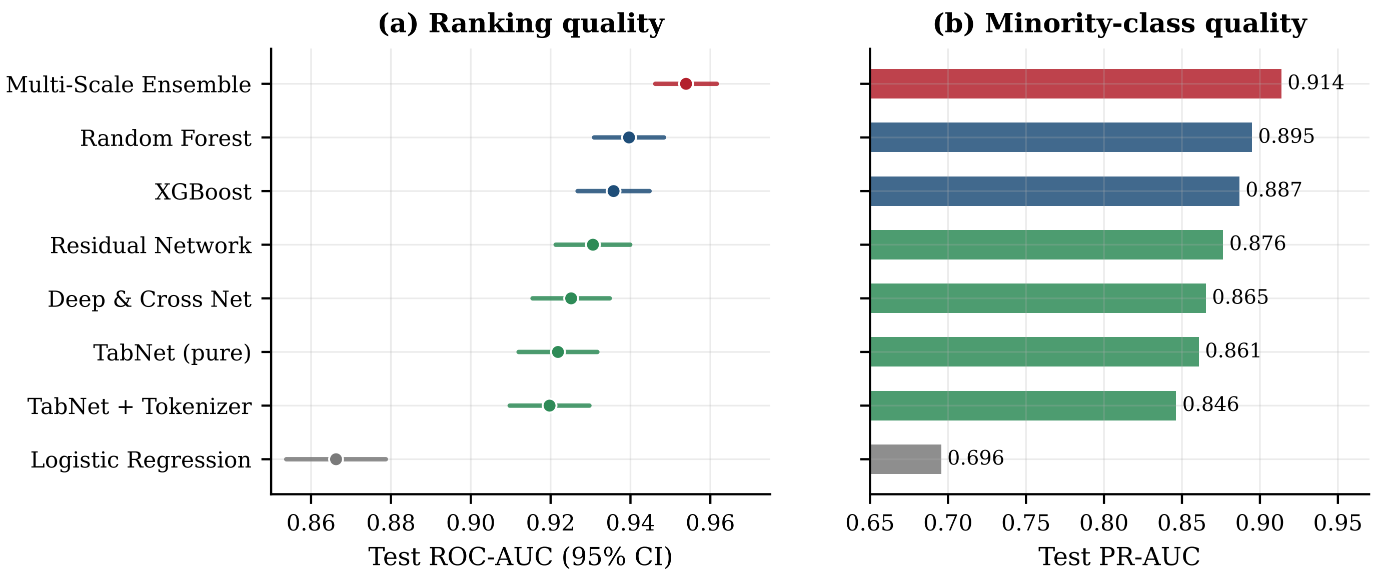

Fig. 3. Test performance with uncertainty. (a) ROC-AUC with Hanley–McNeil 95% intervals. The ensemble's interval is disjoint from those of the neural architectures and overlaps the random forest's only marginally. (b) PR-AUC, the more informative summary under a 3.58:1 imbalance.

TABLE IV
TEST CONFUSION MATRICES AND WILSON 95% INTERVALS AT EACH MODEL'S F_1-OPTIMAL THRESHOLD. n = 6,517 (1,422 DEFAULTS).

Model	TN	FP	FN	TP	Recall (95% CI)	Prec. (95% CI)
Ensemble	5055	40	364	1058	[0.721, 0.766]	[0.951, 0.973]
Random forest	5051	44	370	1052	[0.716, 0.762]	[0.947, 0.970]
XGBoost	5073	22	410	1012	[0.688, 0.735]	[0.968, 0.986]
Residual net	5015	80	401	1021	[0.694, 0.741]	[0.911, 0.941]
Logistic reg.	4525	570	473	949	[0.643, 0.691]	[0.600, 0.649]

A linear model, unable to represent that dependence, must recover the signal from the constituent variables. The ensemble represents the composite directly and the constituents become redundant. The practical consequence is specific and easy to get wrong: a compliance narrative generated from ensemble attributions will not mention prior defaults, even though prior defaults demonstrably influence the outcome through the grade. An adverse-action notice that omits them may be accurate about the model and still misleading about the decision. Attribution to a feature is not attribution to a cause, and collinear composites make the gap operationally visible.

B. Attribution is diffuse, not concentrated

The earlier draft asserted that the top five features account for "almost 75%" of attribution mass. Recomputed from the stored SHAP matrix, the figure is 67.4%; eight of seventeen features are required to reach 80%. Locally the picture is similar: the median instance requires six features to account for 80% of its own $|\phi|$ mass (IQR 5–6).

This matters directly for the narrative layer, which is handed only the top four drivers. Averaged over the explained instances, those four carry 72.3% of an instance's attribution mass, so the prompt discards more than a quarter of the evidence before the language model sees it. The truncation is a defensible engineering choice, since shorter prompts are easier to constrain, but it is a lossy summary and the residual is not negligible.

C. Cross-model agreement is high; SHAP–LIME agreement is not

Two agreement questions are often conflated. They have different answers here.

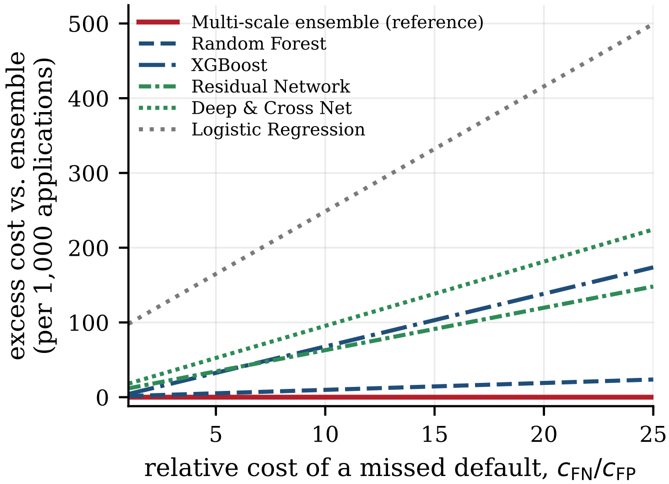

Fig. 4. Excess cost-weighted loss relative to the proposed ensemble at each model's F_1-optimal threshold, as the relative cost of a missed default varies. The ensemble is best throughout, but its margin over a tuned random forest is small: under 2% across the plotted range. The visible separation is between the ensemble and the *neural* models, not between the ensemble and the tree baselines.

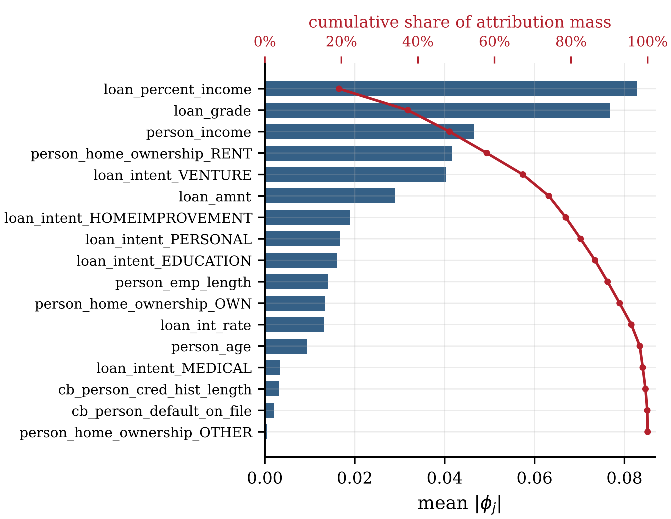

Fig. 5. Global attribution profile of the ensemble over 100 explained test instances (bars, left axis: mean $|\phi_j|$; line, top axis: cumulative share). Attribution is diffuse: the top five features carry 67.4% of the mass and eight are needed to reach 80%. Note the position of `loan_int_rate` (12th) and `cb_person_default_on_file` (16th), both of which are leading predictors in the logistic baseline.

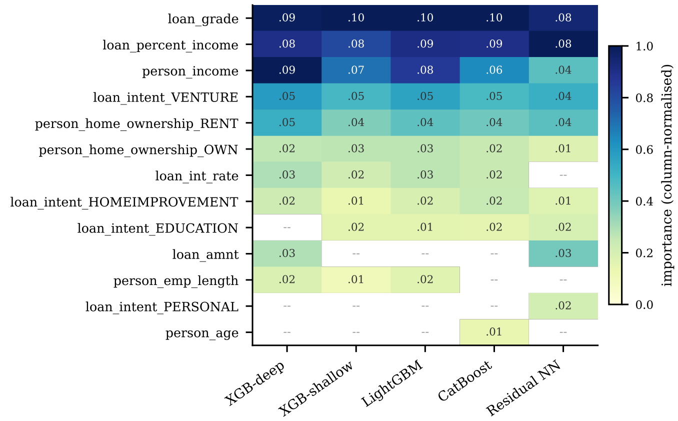

Fig. 6. Per-base-learner SHAP importances, normalised by column maximum; "–" marks a feature outside that learner's top ten. Mean pairwise overlap@10 is 0.86 and mean Kendall τ is 0.78. `loan_grade` and `loan_percent_income` appear in every learner's top three.

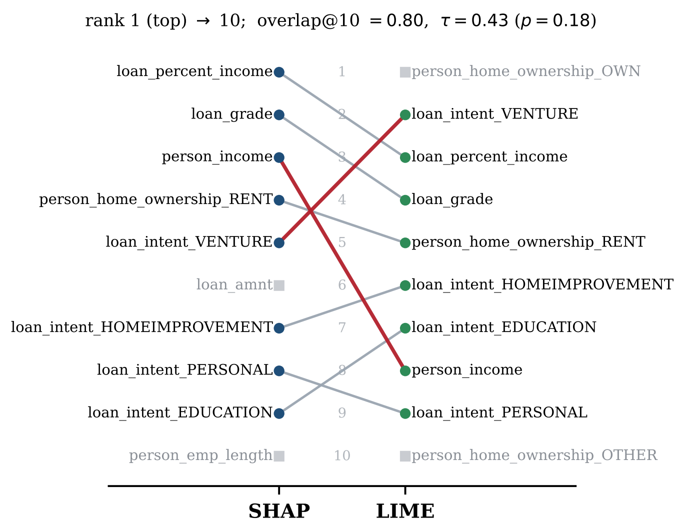

Fig. 7. Global rankings under SHAP and LIME. Grey squares mark features present in only one ranking; red lines mark rank shifts of three or more. The methods largely agree on set membership (overlap@10 = 0.80) and disagree on order ($\tau = 0.43$, $p = 0.18$), including at the very top: LIME's first-ranked feature is SHAP's eleventh.

Across base learners, agreement is genuinely high (Fig. 6). Mean pairwise overlap@10 is 0.860 and mean Kendall τ is 0.776; `loan_grade` and `loan_percent_income` appear in all five learners' top three, and seven features appear in all five top-ten lists. Disagreement is concentrated in low-frequency one-hot levels: `loan_intent_VENTURE` has a coefficient of variation of 0.833 across learners, `cb_person_default_on_file` 0.695. So the claim that the ensemble's headline drivers are architecture-independent is supported.

Across attribution methods, agreement is weaker than the framework assumes (Fig. 7). SHAP and LIME share 8 of 10 top features (overlap@10 = 0.80), but rank correlation on the common set is only $\tau = 0.429$ ($p = 0.179$), Spearman $\rho = 0.524$ ($p = 0.183$), neither significant at $n = 8$. Overlap@3 is 1/3. LIME's top-ranked feature, `person_home_ownership_OWN`, is eleventh under SHAP; SHAP's top feature is LIME's third.

Hypothesis H4 as originally stated, namely that SHAP and LIME show high agreement on top features and that their combination is therefore more robust, is supported for *membership* and unsupported for *ordering*. This is a material qualification, because the fusion step averages normalised scores and then truncates to the top four. Averaging two rankings that disagree on order produces a third ranking that is not the consensus of anything; it is an artefact of the normalisation. Where the methods agree on the set but not the order, a set-level fusion (intersection or union with equal weight) would be more defensible than a score-level average.

D. Attribution sign is unstable for several leading features

A narrative asserts direction: it says a factor raised or lowered risk. That proposition is only meaningful if the attribution's sign is stable enough to support it. Fig. 8(a) measures this as the share of explained instances on which each feature takes its modal sign.

The results are uneven. `loan_percent_income` and `loan_intent_VENTURE` take a consistent sign on 79% of instances and `loan_grade` on 76%, high but not decisive. `person_income` sits at 0.53, statistically indistinguishable from a coin flip; `loan_amnt` at 0.59; `person_home_ownership_RENT` at 0.63.

Some of this is legitimate interaction: income raises risk when paired with a large principal and lowers it otherwise, and a model that captures that is doing its job. But it has a consequence the framework must respect. Any statement of the form "the applicant's income increased their risk" is a claim about one instance that cannot be generalised, and the fusion step, which averages *normalised magnitudes* and discards sign, provides the language model with no signal about how reliable the direction is. Fig. 8(b) sharpens the point: `person_income` is the input to which the ensemble's output is *most* sensitive ($\overline{|\Delta\hat{p}|} = 0.0356$) and simultaneously the one whose attribution direction is least stable.

This is the mechanism we see fail in the next section.

X. Fidelity Audit of the Narrative Layer

A. Why an audit is needed

The pipeline's grounding argument is entirely structural: the prompt contains only computed drivers, the model is told to translate rather than infer, and decoding is deterministic. Determinism is not fidelity. A greedy decode produces the same output every time; it does not produce a *correct* one.

We therefore define a minimal post-generation audit. Given an instance with fused drivers $\{(j, \bar{a}_j)\}$ and generated narrative y:

- **Membership.** Every feature named in y must appear in the driver set supplied in the prompt.
- **Direction.** For every named feature, the direction asserted in y (raises / lowers risk) must match $\text{sign}(\bar{a}_j)$.
- **Coverage.** The prompt's top driver by $|\bar{a}_j|$ should be named.

These are necessary conditions, not sufficient ones, and they are cheap: all three are checkable automatically against data already in memory.

B. An audited failure

Table V reports the audit on the worked case distributed with the reference implementation: an applicant aged 34, income \$92,000, renting, requesting \$92,000 for a stated venture purpose at grade B and 13.5%, with no bureau default. The ensemble returns $\hat{p} = 0.9402$, band HIGH.

TABLE V
FIDELITY AUDIT OF THE WORKED CASE. ATTRIBUTIONS ARE THE FUSED SHAP/LIME IMPACTS SUPPLIED IN THE PROMPT; THE ASSERTION COLUMN IS WHAT THE GENERATED NARRATIVE CLAIMS. THREE OF FOUR CHECKABLE PROPOSITIONS INVERT THE SIGN OF THE EVIDENCE.

Feature	Impact	Evidence	Narrative asserts
`loan_amnt`	+0.3297	raises risk	*not mentioned*
`..._ownership_RENT`	+0.1179	raises risk	raises risk ✓
`..._ownership_OWN`	+0.0815	raises risk	*not mentioned*
`loan_intent_VENTURE`	−0.0720	**lowers** risk	raises risk ✗
`person_income`	−0.0569	**lowers** risk	raises risk ✗
`person_age`	−0.0092	**lowers** risk	raises risk ✗

The generated narrative reads:

> *"The risk level for this borrower is high due to their age, income, and home ownership status. Their intent to take out a venture loan also increases the risk. This case is similar to other high-risk cases in terms of income and home ownership status. However, the intent to take out a venture loan is a unique factor that sets this case apart. Based on these factors, it is recommended to closely monitor this borrower's credit history and consider additional risk mitigation measures."*

It is fluent, appropriately hedged, correctly identifies the risk band, and would pass casual review. It is also wrong in a specific and consequential way.

Three failures are visible in Table V:

1) **Sign inversion.** The narrative asserts that age, income and venture intent *increase* risk. The attributions score all three as risk-*reducing* (−0.0092, −0.0569, −0.0720). The model did not misjudge magnitude; it reversed direction, and did so for the majority of the features it chose to name.
2) **Coverage failure.** The dominant driver, `loan_amnt` at +0.3297, is never mentioned. It carries roughly three times the next largest impact and is the one factor that plainly explains a 94% default probability on a loan equal to annual income.
3) **Membership violation.** `person_age` is asserted as a risk factor although it was not among the four drivers supplied.

The narrative additionally attributes salience to venture intent ("a unique factor that sets this case apart") that the evidence assigns to principal size.

C. Diagnosis

Three properties of the surrounding system make this outcome likely rather than accidental.

The prompt discards sign reliability. Fusion averages normalised magnitudes. The language model receives a signed impact but nothing about how stable that sign is, and Section IX shows that for `person_income` it is barely stable at all (0.53).

Prior knowledge overrides supplied evidence. A 2.7B model has strong priors about credit: high income is good, venture lending is speculative, young borrowers are risky. When the supplied attributions contradict those priors, as they

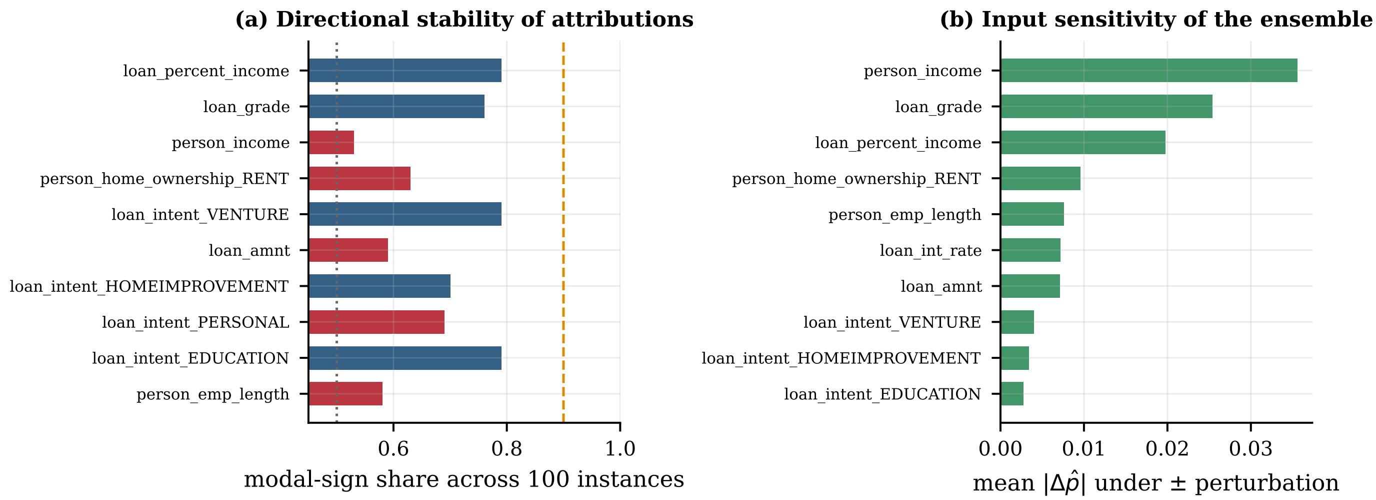

Fig. 8. (a) Directional stability: share of the 100 explained instances on which a feature's attribution takes its modal sign. Bars below 0.7 (red) indicate features whose risk direction routinely flips across applicants; a value near 0.5 means the direction is near-arbitrary at the population level. (b) Sensitivity of the ensemble's output to bounded perturbation of each input. The two panels disagree about `person_income`, which is the most sensitive input yet the least directionally stable.

do here since *this* applicant's income and stated purpose reduce risk relative to the population, the prior wins. The instruction to translate rather than infer does not bind.

Nothing checks the output. Every control is upstream. The system has no mechanism that could have caught a sign inversion, which is why a deterministic decode produced a reproducibly wrong narrative rather than an occasionally wrong one.

D. Scope of this finding

We are explicit about what this is and is not. It is a documented, reproducible failure of a specific model under a specific prompt on a specific instance, verified against the attributions supplied in the same run. It is *not* an estimate of failure frequency: producing one would require generating and adjudicating narratives across a labelled sample, which is future work we describe in Section XIII. A single audited failure cannot establish a rate. It is, however, sufficient to refute the claim the earlier draft made, that the narratives "accurately reflect the underlying feature contributions", and sufficient to establish that input-side constraints alone do not guarantee grounding.

Our recommendation follows directly from the audit definition: the three checks above are cheap, automatic, and would have blocked this output. A narrative layer in a regulated setting should run them and fall back to a deterministic template when they fail. This is the two-stage generate-then-verify pattern argued for in [12], and our result is a concrete instance of why the second stage is not optional.

XI. Trustworthiness Diagnostics

Ranking quality is one property of a scoring model. A credit system uses the probability itself: to price risk, to set the HIGH/MODERATE/ LOW band that appears in the narrative prompt, and to aggregate expected loss. Table VI evaluates the ensemble against the acceptance thresholds our own protocol declares.

TABLE VI
Trustworthiness diagnostics against the acceptance thresholds specified by our own evaluation protocol. Three of four fail.

Diagnostic	Value	Threshold	Verdict
Calibration (ECS, weighted)	0.1171	< 0.05	fail
unweighted	0.2262	n/a	n/a
Perturbation stability (DPD)	0.0780	< 0.05	fail
Local explanation consistency	$\overline{\mathrm{CV}} = 1.1495$	n/a	low
Group disparity (EOD)	0.0640	< 0.10	pass

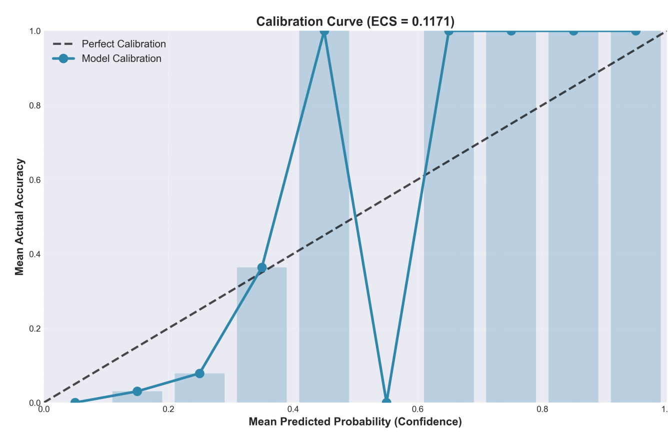

Fig. 9. Reliability diagram for the ensemble ($n = 500$, 10 bins, ECS $=$ 0.1171). The model is over-confident below $\hat{p} \approx 0.3$ and erratic above it. The upper bins are sparsely populated, so the extreme excursions should be read as high-variance rather than as evidence of systematic bias. The low-probability region, where most applications fall, is well populated and consistently miscalibrated.

Calibration fails. The expected calibration score is 0.1171 weighted and 0.2262 unweighted, against a 0.05 acceptance threshold. Fig. 9 shows systematic over-confidence in the low-probability region where the overwhelming majority of applications sit. This is the expected consequence of two design choices: focal loss deliberately distorts the probability scale in exchange for better minority-class ranking, and modern neural networks are miscalibrated by default [39]. The remedy is standard and cheap, namely Platt scaling or isotonic regression on a held-out fold [40], and we did not apply it. We flag this rather than omit it because miscalibrated probabilities propagate directly into the risk band shown to the loan officer.

Perturbation stability fails. Under bounded input perturbation (strength 0.01, 10 repetitions) the mean absolute prediction shift is 0.0780 ($\sigma = 0.0595$, max 0.3284), against a 0.05 threshold. A 1% input perturbation moving the output by nearly 8 points is a material sensitivity for a system whose inputs are self-reported.

Local explanation consistency is low. Clustering the explained instances into five groups and measuring within-cluster dispersion of attribution vectors gives a mean coefficient of variation of 1.1495, so dispersion exceeds the mean. Similar applicants receive materially different explanations. This is consistent with the sign instability of Section IX and with LIME's documented sampling variance [17].

Group disparity passes, with an important caveat. Equalised odds difference [41] is 0.0640 overall, below the 0.10 threshold, with the largest gap on the low-debt-ratio split (0.0839). The caveat is that the groups examined (grade presence, high income, low debt ratio) are *model-derived proxies*, not protected attributes. The dataset contains no race, gender or ethnicity fields, so no statement about discrimination on protected characteristics can be made from these numbers, and we make none. Reporting a passing EOD without that qualification would be the most misleading thing in this paper.

XII. Discussion

A. Where the gains are, and are not

Read together, our results describe a system whose two halves succeed differently.

The predictive half works, modestly. Multi-scale stacking beats the best single model by a margin that survives a deliberately unfavourable significance test, generalises without overfitting despite six trainable components, and improves PR-AUC by more than it improves ROC-AUC, which is the right direction under imbalance. But its operational advantage at the deployed threshold is six defaults in six thousand decisions. An institution weighing five base learners, a meta-network and the attendant governance burden against a single random forest should know that number before committing.

The explanatory half is where the interesting failure is, and it is not where the literature usually looks. The attribution layer is sound in construction: explaining the composed pipeline rather than a component is the right choice, and cross-learner agreement confirms the headline drivers are not artefacts. What breaks is the last step, the one added specifically to make the system usable by non-specialists. That step inherits every instability upstream of it (diffuse attribution, unstable signs, method disagreement on ordering), compresses them into four numbers, and hands them to a model with strong independent priors and no obligation to defer.

B. Implications for LLM-based explanation interfaces

Three points generalise beyond this system.

Constrained prompting is a necessary but not sufficient control. Our prompt does everything the literature recommends. It still produced sign inversions. The reason is structural: prompt constraints shape the input distribution and cannot bind the output. Any deployment that treats prompt design as the safety mechanism is relying on a control that has no enforcement.

Verification is cheap and should be mandatory. Membership, direction and coverage checks require no model calls, no labels and no human review. They would have blocked our failure case. In a regulated setting, an unverified narrative is a liability whose cost is bounded by the size of the credit decision, and the check costs microseconds.

Explanation quality inherits attribution quality. A narrative cannot be more faithful than the attributions it summarises. Reporting sign stability and inter-method rank agreement alongside importance rankings should be standard, because these properties, not mean $|\phi|$, determine whether a directional claim about an individual applicant is supportable.

C. Practical guidance

For practitioners building similar systems, our results suggest an ordering of effort that differs from the one we followed. Calibrate before deploying, since miscalibrated probabilities corrupt both pricing and the risk bands that reach human reviewers, and the fix is a single held-out fold. Optimise the threshold against an explicit cost ratio rather than F_1; Fig. 4 indicates this matters more than the choice among strong models. Verify narratives after generation. And treat additional base learners as the last resort rather than the first, since our fifth learner bought less than our threshold choice did.

XIII. Limitations and Future Work

Single dataset. All results come from one public dataset with a fixed 21.8% default rate, no temporal structure and no macroeconomic covariates. Whether the meta-learner's weighting generalises across imbalance regimes, feature distributions or credit cycles is untested.

The fidelity audit is a case study. We document one reproducible failure in detail. We do not estimate a failure rate. The natural extension is to generate narratives for a stratified sample of several hundred test instances and score them automatically against the three audit criteria, which would yield per-criterion violation rates and support comparison across model scales and prompt designs. We regard this as the single highest-value follow-up and note that our audit definition makes it inexpensive.

Correlated AUC comparison. Paired score vectors were not retained, so DeLong's test [43] could not be run. Our conservative substitute understates significance; re-running with retained scores would tighten the comparison in our favour, which is the direction that requires no defence.

No calibration stage. We report miscalibration but do not correct it. Adding Platt or isotonic scaling on a dedicated fold, then re-deriving the risk bands, is straightforward and should precede any deployment.

Fixed base hyperparameters. Base configurations were set from prior experiments and held fixed while the meta-learner was tuned. Joint optimisation might improve the ensemble, at substantially higher cost.

Static evaluation. Batch prediction on a fixed split does not test concept drift, which is the dominant failure mode of deployed credit models across economic cycles.

Protected attributes absent. The dataset contains no protected characteristics, so our disparity analysis uses proxies and cannot speak to discrimination.

Beyond these, two directions follow naturally from our findings: fine-tuning or constrained decoding that forces the narrative model to respect supplied signs, and set-level rather than score-level attribution fusion, which Section IX suggests would be better founded given that the two methods agree on membership but not order.

XIV. Conclusion

We built a credit-risk system of the kind currently advocated in the explainable-AI literature: a high-performing ensemble, model-agnostic attribution over the composed pipeline, and a constrained language model to render the attributions as prose. We then measured both halves against their own claims.

The ensemble performs as intended. Multi-scale stacking of four differently-regularised boosting learners and a residual network reaches test ROC-AUC 0.9539 and PR-AUC 0.9137, improving on the best single model by a margin that remains significant ($p = 0.016$) under a test chosen to disadvantage it, with no evidence of overfitting. We also show that this margin is worth less operationally than it appears: under 2% of cost-weighted loss at the deployed threshold.

The narrative layer does not perform as claimed. Constrained prompting, deterministic decoding and top-k truncation did not prevent a generated rationale from inverting the risk direction of three of the four factors it named, omitting the dominant driver, and introducing a feature it was never given. We trace this to measurable properties of the attribution layer beneath it: attribution is diffuse (67.4% mass in five of seventeen features), SHAP and LIME agree on membership but not order ($\tau = 0.43$, $p = 0.18$), and the sign of the attribution for the ensemble's most sensitive input is near-random across applicants (0.53). We report alongside these that the ensemble is miscalibrated ($\text{ECS} = 0.117$) and sensitive to bounded input perturbation ($\text{DPD} = 0.078$), both outside our own acceptance thresholds.

The conclusion we draw is not that language models have no place in credit explanation. It is that grounding must be verified rather than assumed, that the verification is cheap, requiring only membership, direction and coverage checks over data already in memory, and that a system which enforces it can make the claim we could not: that the explanation given to the applicant matches the evidence the model actually used.

Appendix A
Narrative Prompt Template

The template below is reproduced verbatim from the deployed service. Bracketed italics denote substituted values.

```
You are an AI credit risk officer. Provide a
clear, professional explanation.
Prediction: [p]% default probability → Risk
Level: [band]
Borrower Profile:
[raw feature: value, one per line]
Local Feature Drivers (for this specific case):
[feature: increases/decreases risk (impact:
±x.xxx), top 4]
Global Context: Top globally important features
are [top 5].
Most sensitive features (from analysis): [top 3].
Provide a 2-3 sentence explanation: (1) State the
risk level and key factors, (2) Explain how this
case compares to typical patterns, (3) Give a
recommendation.
```

Generation uses the following decoding parameters.

Parameter	Value
`max_new_tokens`	256
`min_new_tokens`	64
`do_sample`	`False` (greedy)
`repetition_penalty`	1.05

The driver list carries signed impacts, so the direction that the audit in Section X finds inverted was explicitly present in the prompt.

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